# LAB 03 ANSWERS REPORT

HANS PUJALTE (PUJA0009)

COMP3742, TONSLEY
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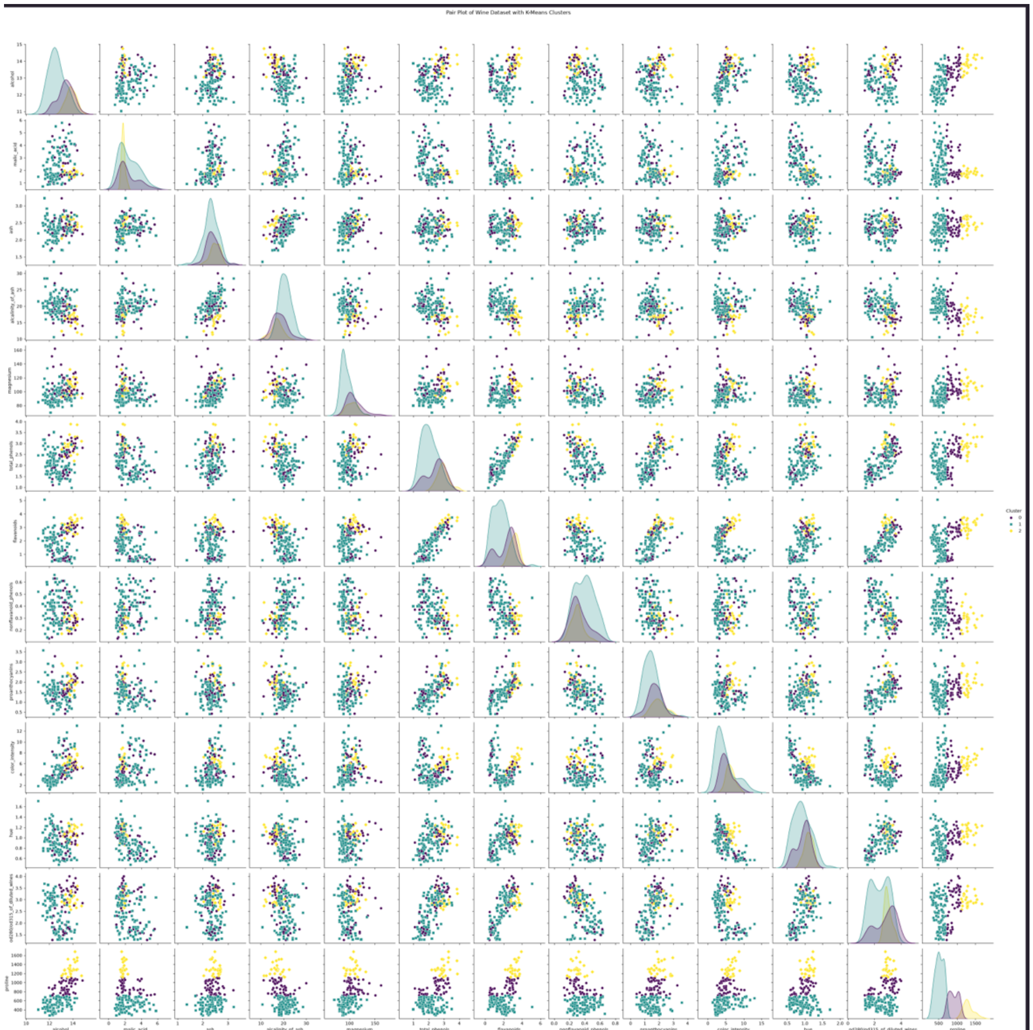
Hans Pujalte

1. **Apply K-Means clustering to the Wine dataset and visualize the clusters using a 2D scatter plot.**




2. **Calculate the silhouette score for the K-Means clustering performed on the Wine dataset**

The average silhouette score for the K-Means clustering was 0.514.

3. **Apply K-Means clustering to the Wine dataset, determine the optimal number of clusters using the Elbow method, and visualize the clusters via plots. Compare the results obtained with those found using the silhouette score.**

Using the Elbow method, the optimal number of clusters was 2 with a silhouettes score of 0.65. In contrast, the average silhouette score was 0.514. It is important to note that the data is not scaled for this question.

4. **Perform K-Means clustering on the Wine dataset, determine the optimal number of clusters using the Elbow method, and then use cluster centroids to classify new, unseen data points. Evaluate the classification accuracy and discuss any potential improvements.**

Using the Elbow method and scaling the data, the optimal number of clusters identified was 3. After mapping the cluster IDs to the wine classes, the model achieved an accuracy of approx. 97%. This high accuracy suggest that the identified cluster centroid positions correspond closely to the identified wine classes. A possible improvement could be to utilise feature scaling since features with larger numerical range could dominate/influence the distance calculations using in cluster formatting and classification.

5. **Apply K-Means clustering to the Wine dataset. Experiment with different feature scaling methods (e.g., StandardScaler, MinMaxScaler) and KMeans initialization techniques (e.g., k-means++, random). Compare the results using silhouette scores and visualize the clusters using pair plots.**

For this question, the Standard, MinMax and Robust scaler was used for experimentation. For the initialisations, k-means++ and random were used. MinMax scaler achieved the highest silhouette score across all scalers and for both initialisations. Interestingly, the difference in silhouette scores between k-means ++ and random was minor. The clusters were then visualised using pair plots (refer to the repo). For simplicity, 5 features were used to generate the pair plot.